CLOUD SECURITY — PRACTICAL-BASED LEARNING

PBL 2: Deploy a Public Web Server on a Windows EC2 Instance

Standalone practical report

1. Aim

Deploy a Windows-based Amazon EC2 web server using IIS and publish a webpage displaying the student's last four PRN digits.

2. Problem Statement

Launch a Windows EC2 instance, connect using Remote Desktop Protocol (RDP), install and configure IIS, create the required webpage, allow HTTP access, and verify the page using the instance's public IP address.

3. Tools and Technologies

- AWS Management Console and Amazon EC2
- Windows Server EC2 image
- Remote Desktop Protocol (RDP)
- Internet Information Services (IIS)
- EC2 key pair and Administrator password retrieval
- Security Groups and HTML

4. Recorded Instance Details

Historical details from troubleshooting: instance name PBL2-Web-Instance; instance ID i-0fe795bba5770c93e; public IP 15.207.117.5; private IP 172.31.3.236. These are historical values and must be checked in the AWS console before reuse; public IPs may change.

5. Procedure

Step 1 — Launch a Windows EC2 instance

Launch a Windows Server instance and select or create a key pair. Wait for the instance to become ready and its status checks to pass.

Step 2 — Configure the security group

- Allow RDP (TCP 3389) only from the authorized client public IP/32.
- Allow HTTP (TCP 80) for the intended website access.

Step 3 — Retrieve Administrator credentials

Use the AWS EC2 password retrieval workflow with the matching private key to decrypt the initial Administrator password. Use the key pair associated with that instance.

Step 4 — Connect using RDP

Open Remote Desktop Connection, enter the instance's current public IP, and authenticate as Administrator using the retrieved password.

Step 5 — Install IIS

Use Server Manager to add the Web Server (IIS) role and complete the installation. Confirm the IIS service is running.

Step 6 — Create the webpage

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C:\\inetpub\\wwwroot\\index.html
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Create or edit the default HTML page in the IIS web root and include the correct last four digits of the PRN.

Step 7 — Test HTTP access

Open http://PUBLIC_IP in a browser and verify the page displays the expected content. Confirm the public IP and security-group rules if it fails.

6. Errors Encountered and Solutions

EC2 status checks showed 0/2

The instance initially showed failed or incomplete status checks. Wait for initialization and inspect EC2 system and instance status checks. If failures persist, inspect instance state, system logs and console output before retrying RDP.

Systems Manager showed disconnected

The instance appeared disconnected in Systems Manager. Systems Manager access depends on agent status, IAM permissions and network connectivity. This status alone does not establish whether RDP is working; use the appropriate RDP workflow for this practical.

Administrator password could not be retrieved or accepted

Use the private key that matches the instance's selected key pair and ensure the instance has completed initialization. Verify the username and keyboard/input settings. Do not publish the decrypted password.

RDP connection problem

Check that the instance is running, status checks have passed, the current public IP is correct, and TCP 3389 is allowed only from the authorized client IP. Verify the instance has a public route and that local firewall/network restrictions are not blocking RDP.

HTML formatting or saving issue

Save the file as index.html in C:\\inetpub\\wwwroot, not as a text file with an additional .txt extension. Reopen the file to confirm the HTML content and refresh the browser.

Website inaccessible

Confirm IIS is installed and running, the file exists in the IIS web root, and the security group allows inbound TCP 80. Verify using the current public IP. The available history does not confirm a final successful external browser test.

7. Result

The practical was intended to publish a public website using IIS on Windows EC2. Troubleshooting included initial EC2 status-check issues, Systems Manager disconnection, Administrator password/RDP difficulties and HTML file handling.

FINAL STATUS: NOT FULLY VERIFIED. The available history does not prove that an external browser successfully displayed the webpage with the correct PRN suffix. Capture that evidence before marking the deployment complete.

8. Security Measures

- Restrict RDP to the authorized client public IP/32; do not open port 3389 to everyone.
- Allow HTTP only as required by the assignment.
- Keep Administrator credentials and the EC2 private key confidential.
- Use a supported Windows image and install only required roles.
- Monitor AWS costs and stop or delete the instance after the practical.

9. Conclusion

This practical covers launching a Windows EC2 instance, retrieving its initial Administrator password, connecting over RDP, installing IIS and publishing a static webpage. Several access and configuration issues were investigated, but a successful public browser test remains unverified in the available history.

10. Screenshots Checklist

- ■ Windows EC2 instance details and status checks.
- ■ Security group with RDP restricted to authorized public IP/32 and HTTP port 80 allowed.
- ■ Administrator password retrieval screen without exposing the password.
- ■ Successful RDP desktop.
- ■ IIS role installed and service running.
- ■ index.html saved in C:\\inetpub\\wwwroot.
- ■ External browser showing the correct PRN suffix.
- ■ Resource cleanup evidence.